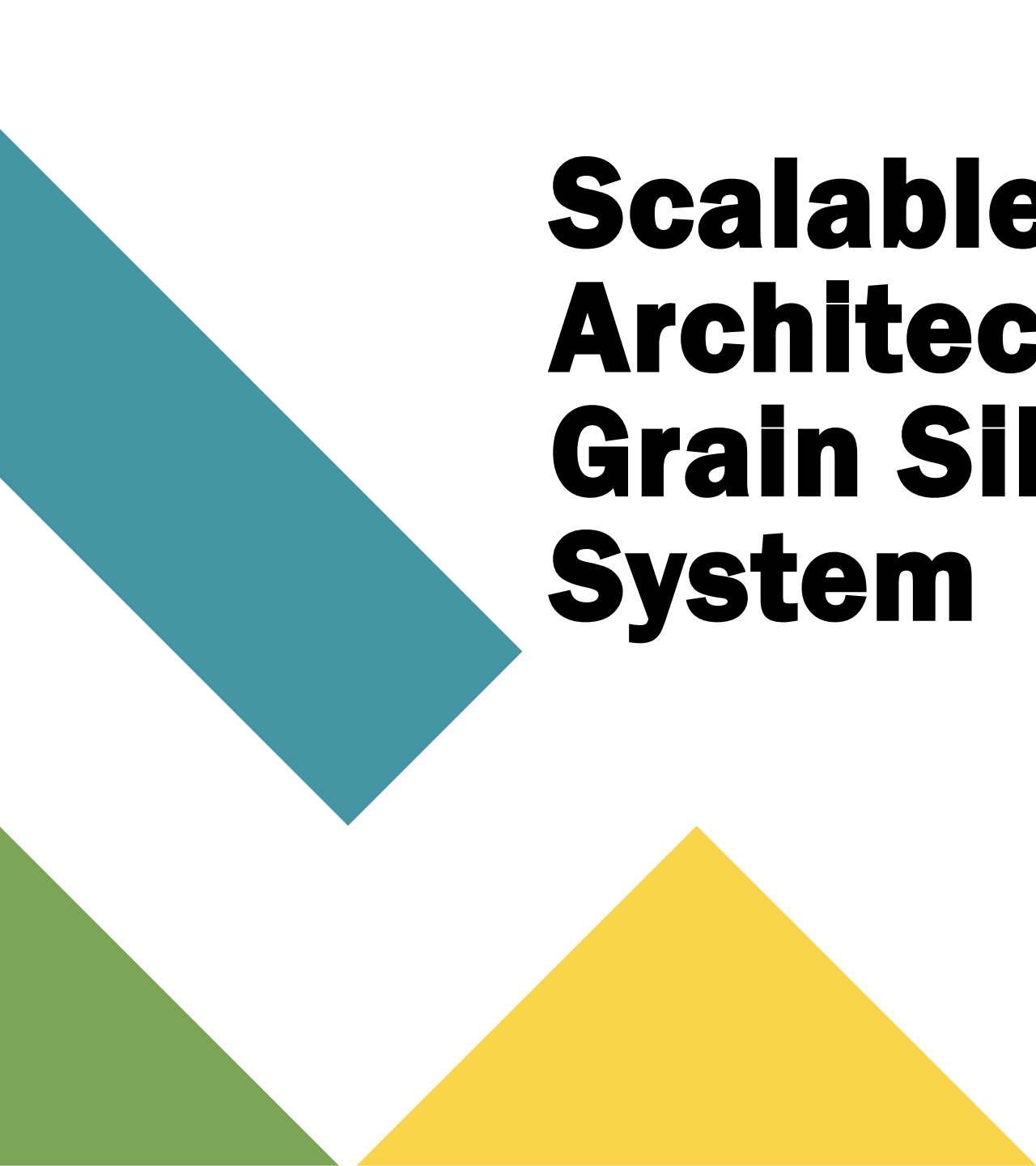




# **Scalable Fog–Cloud Architecture for Smart Grain Silo Monitoring System**

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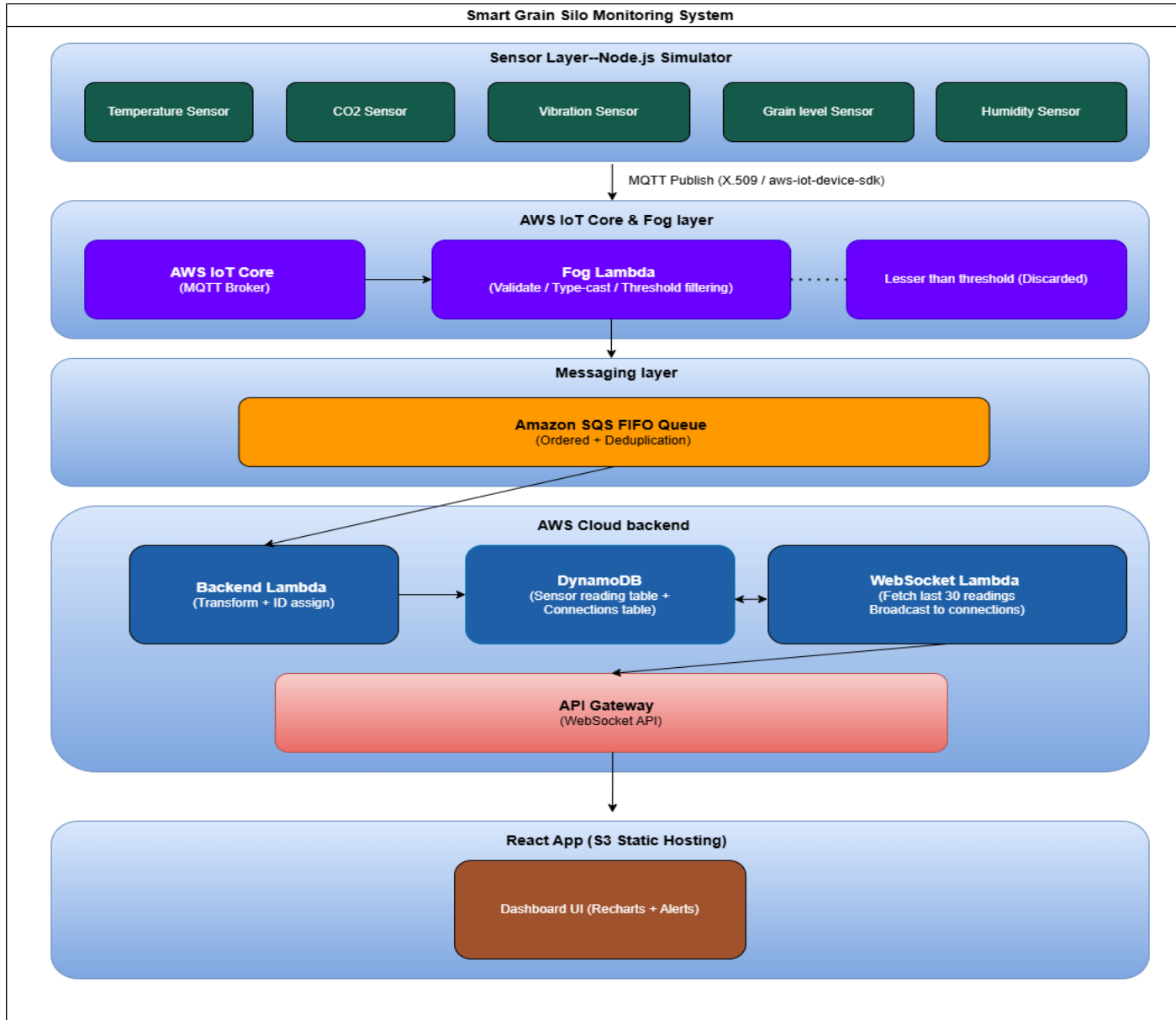
Subin George  
Student ID: 24111945

# What is a Grain Silo & Why Monitor It?



- A grain silo is a large storage structure used to store harvested grains such as wheat, corn, rice, and barley over long periods of time.
- **Temperature & Humidity** — Poor conditions accelerate grain spoilage and encourage fungal and microbial growth.
- **CO<sub>2</sub> Buildup** — Rising CO<sub>2</sub> levels inside the silo signal anaerobic fermentation or pest infestation.
- **Vibration** — Unusual vibrations indicate structural stress or mechanical failure risks.
- **Grain Level** — Unexpected drops in grain level may signal a leak or breach, enabling early intervention and preventing grain loss.

# System Architecture



- **Sensor Layer:** Node.js Simulator  
MQTT / AWS IoT Core
- **Fog Layer:** AWS Lambda Filtering & Thresholds
- **Messaging Layer:** Amazon SQS FIFO  
Ordering & Deduplication
- **Backend Layer:** Lambda + DynamoDB  
Storage & Processing
- **Presentation Layer:** React Dashboard  
WebSocket API

# Sensor Parameters & Thresholds

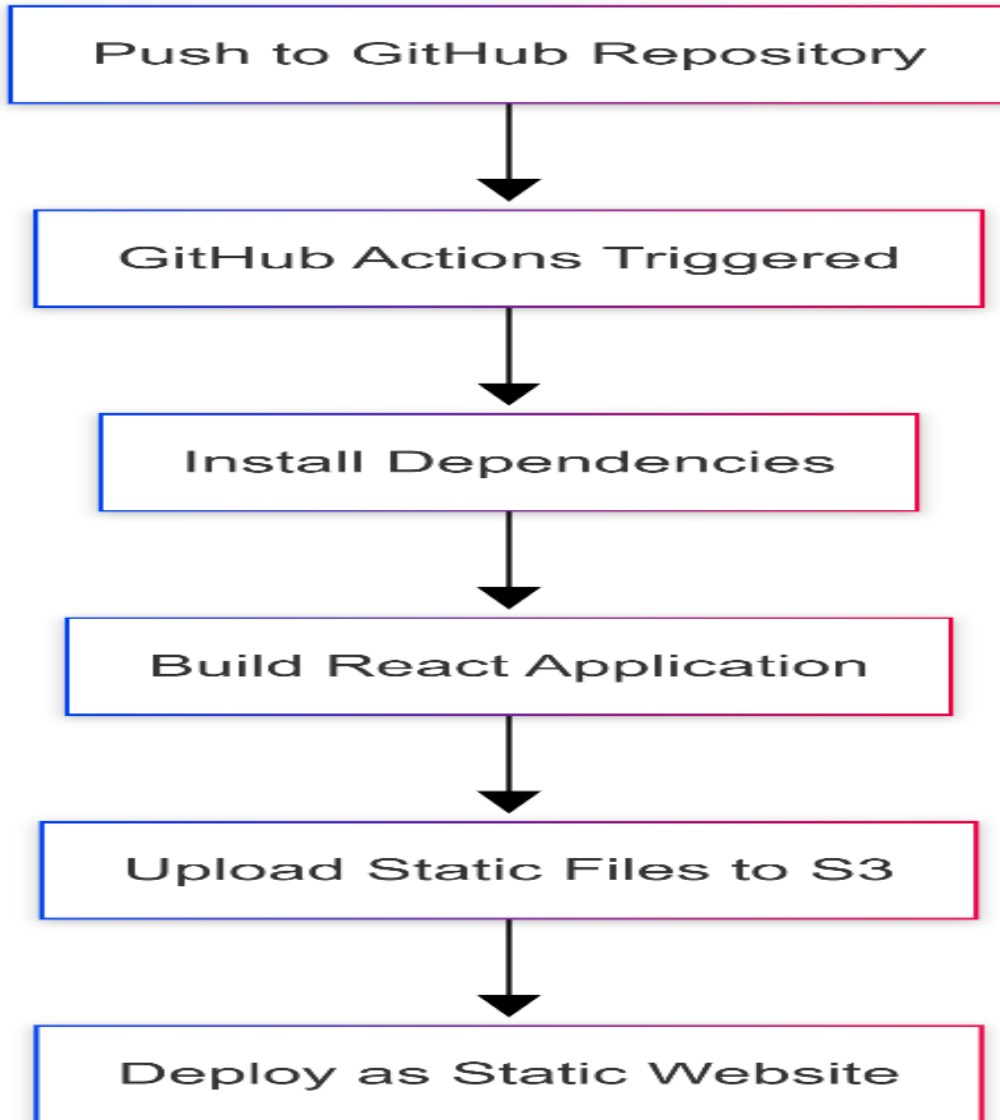
| Parameter             | Range          | Alert Threshold | Real-World Sensor |
|-----------------------|----------------|-----------------|-------------------|
| Temperature (°C)      | 25 – 40 °C     | > 30 °C         | PT100 / DS18B20   |
| Humidity (%)          | 50 – 80 %      | > 60 %          | SHT31 / DHT22     |
| CO <sub>2</sub> (ppm) | 400 – 1500 ppm | > 1000 ppm      | MH-Z19 / SCD30    |
| Vibration (g)         | 0 – 1.5 g      | > 0.5 g         | ADXL345 / MPU6050 |
| Grain Level (%)       | 50 – 100 %     | < 70 %          | HC-SR04 / HX711   |

# Design Patterns & Scalability

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- **Serverless Computing** — AWS Lambda auto-scales with workload fluctuations; pay-per-use keeps costs low.
- **Event-Driven Architecture** — Components communicate via events (MQTT messages, SQS triggers), keeping the system loosely coupled and flexible.
- **Publish-Subscribe Messaging** — Reducing communication overhead and enabling scalable data distribution.
- **Queue-Based Load Levelling** — Amazon SQS absorbs data spikes, preventing overload and ensuring fault-tolerant, steady processing.
- **Real-Time Push via WebSockets** — Bidirectional client-server communication delivers live updates to the dashboard with minimal network overhead.
- **DynamoDB for Scalable Storage** — NoSQL database with automatic scaling, high throughput, and low-latency reads; no schema management required, ideal for variable IoT data.

# CI/CD Pipeline



**Push to GitHub Repository** — push updated code to the GitHub repository, triggering the automated pipeline.

**GitHub Actions Triggered** — The workflow is automatically activated on every push

**Install Dependencies** — Pipeline installs all required Node.js packages and libraries needed to build the React application.

**Build React Application** — The React app is compiled into optimised static files (HTML, CSS, JavaScript) ready for production.

**Upload Static Files to S3** — The generated build files are uploaded to an Amazon S3 bucket configured for static website hosting.

**Deploy as Static Website** — S3 serves the files as a publicly accessible static website

# Live Demo

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- <http://smart-grain-silo-monitoring-system.s3-website-us-east-1.amazonaws.com/>

**Thank you**



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